Systematic Review of the Incidence of Sudden Cardiac Death in the United States

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The need for consistent and current data describing the true incidence of SCA and/or SCD was highlighted during the most recent Sudden Cardiac Arrest Thought Leadership Alliance's (SCATLA) Think Tank meeting of national experts with broad representation of key stakeholders including thought leaders and representatives from the American College of Cardiology, American Heart Association, and the Heart Rhythm Society. As such, to evaluate the true magnitude of this public health problem, we performed a systematic literature search in MEDLINE using the MeSH headings, "death, sudden" OR the terms "sudden cardiac death" OR "sudden cardiac arrest" OR "cardiac arrest" OR "cardiac death" OR "sudden death" OR "arrhythmic death." Study selection criteria included peer-reviewed publications of primary data used to estimate SCD incidence in the U.S. We used Web of Science®'s Cited Reference Search to evaluate the impact of each primary estimate on the medical literature by determining the number of times each "primary source" has been cited. The estimated U.S. annual incidence of SCD varied widely from 180,000 to > 450,000 among 6 included studies. These different estimates were in part due to different data sources (with data age ranging from 1980 to 2007), definitions of SCD, case ascertainment criteria, methods of estimation/extrapolation, and sources of case ascertainment. The true incidence of SCA and/or SCD in the U.S. remains unclear with a wide range in the available estimates, which are badly dated. As reliable estimates of SCD incidence are important for improving risk stratification and prevention, future efforts are clearly needed to establish uniform definitions of SCA and SCD and then to prospectively and precisely capture cases of SCA and SCD in the overall U.S. population.